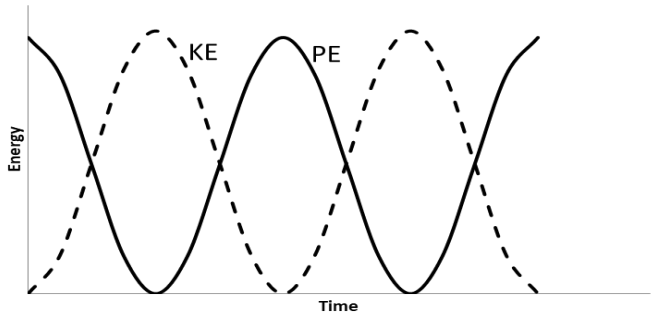


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		Straight line <u>through the origin</u> [shows that acceleration is directly proportional to displacement] (1) Negative slope [shows that the acceleration is always directed towards the fixed point] (1)	2			2		
		(ii)		Calculating gradient i.e. 60 or obtaining suitable point from graph e.g. -0.05, 3.0 (1) Gradient = [-] ω^2 or substitution into equation (1) $\omega = 7.75 \text{ [rad s}^{-1}\text{]}$ (1) Allow 1 mark for slips such as: $\omega = \sqrt{\frac{3}{0.5}}$ Allow 2 marks for $\omega = \sqrt{\frac{3}{0.5}} = 2.45$ or $\omega = \sqrt{\frac{1.5}{0.03}} = 7.07$ Allow 3 marks for Accept $\sqrt{\frac{1.75}{0.03}} = 7.64 \text{ [rad s}^{-1}\text{]}$ or $\sqrt{\frac{2}{0.033}} = 7.8 \text{ [rad s}^{-1}\text{]}$ or $\sqrt{\frac{2.4}{0.04}} = 7.85 \text{ [rad s}^{-1}\text{]}$ Allow 1 mark for: $\omega = \sqrt{\frac{a_{\text{max}}}{A}} = 7.7$ i.e. no evidence of data points used or substitution but method ok. Alternative for full marks: Using $a = \frac{v^2}{r}$ (which leads to $v = 0.39$) (1) Using $\omega = \frac{v}{r}$ (1) Convincing correct answer e.g. 7.75 or $\frac{0.39}{0.05}$ (1)		3		3	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)	I.	$T = \frac{2\pi}{\omega} = \frac{2\pi}{7.7} = 0.82 \text{ [s]}$		1		1	1	
			II.	Substitution: $T = 2\pi\sqrt{\frac{m}{k}}$ so $0.82 = 2\pi\sqrt{\frac{0.30}{k}}$ or $ma = kx$ (1) $k = 17.8 \text{ N m}^{-1}$ unit mark (1)	1	1		2	2	
			III.	Substitution: $v = \omega A = 7.7 \times 0.05$ (1) if $\sin \omega t$ or $\cos \omega t$ used then the angle must be correct $v = 0.39 \text{ [m s}^{-1}\text{]}$ (1) Alternatives for the first mark: Substituting into $ma = kx$ e.g. $0.3 \times 3 = k \times 0.05$ (1) OR substituting into $\omega = \sqrt{\frac{k}{m}}$ e.g. $7.8 = \sqrt{\frac{k}{0.3}}$ (1)	1	1		2	2	
		(iv)		 Shape and start point of PE sketch (1) Shape KE sketch and starting at zero (1) KE out of phase with PE and same size peaks and one complete cycle i.e. 2 KE peaks (1)		3		3	1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			Indicative content: <u>Damping:</u> Damping linked to resistive force. It dissipates energy or linked to energy loss. Amplitude decreases with time. Link to energy loss to surroundings. Critical damping. <u>Resonance:</u> Periodic force. Happens when frequency is equal to the natural frequency. Amplitude of oscillation becomes large. Damping lowers the resonance peak. Reasonable resonance graph. <u>Real systems:</u> Any sensible real system e.g. bridges / structural /swings, tuning circuit Example of damping e.g. car suspension With a description of the driving force Explanation of whether resonance is good or bad.	6			6		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				5-6 marks All 3 areas answered well. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks 2 areas answered well or limited answers to all 3 areas. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i> 1-2 marks 1 area answered well or limited answers to 2 areas. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
				Question 3 total	10	9	0	19	8	0